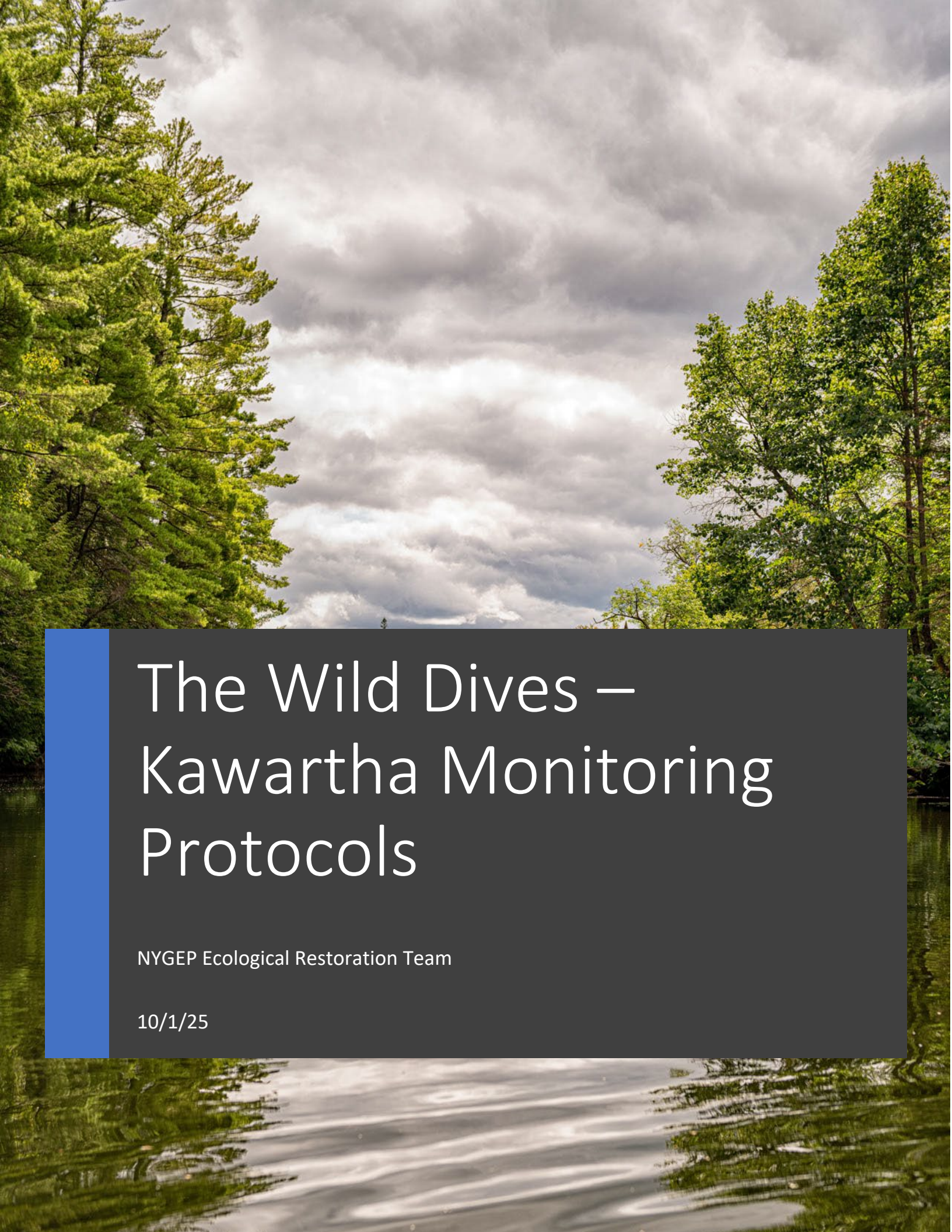




The Wild Dives – Kawartha Monitoring Protocols

NYGEP Ecological Restoration Team

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Introduction

As part of The Wild Dives – First Descent Geographic Expeditions Project launched by the Toronto and Region Conservation Authority (TRCA), the Ecological Restoration team from the Newcomer Youth Green Economy Project (NYGEP) will be responsible for creating monitoring protocols to help achieve the project goals in Kawartha Highlands Provincial Park, specifically in the Kawartha Highland Signature Site Park (fig. 1). Excluding Algonquin Park that extends into northern Ontario, Kawartha is the largest single area of preserved land in Southern Ontario spanning across 37,587 ha (Ontario Parks, 2025).

Kawartha Highlands Provincial Park protects a highly diverse natural area straddling the boundary of two ecoregions. Kawartha Highlands is a large, relatively intact natural area and is part of a forest landscape that joins major protected areas to the north (Algonquin Provincial Park and Silent Lake Provincial Park), the east (Petroglyphs Provincial Park, the Peterborough Crown Game Preserve, and the Sharpe Bay Fen Conservation Reserve), the west (Queen Elizabeth II Wildlands and Kawartha Barrens Enhanced Management Area), and the south (Wolf Island Provincial Park). Its extensive forests, rock barrens, lakes, rivers, and wetlands provide diverse habitat for at least 37 mammal species (53% of the provincial total), 176 bird species (146 confirmed breeding), 13 reptiles, 16 amphibians, 74 dragonflies and damselflies, 65 butterflies, and 688 vascular plants, of which 628 are native (Ontario Parks, 2008).

Many remote freshwater ecosystems remain undocumented, especially those beneath the surface. For all three targeted sites (Algonquin, Kawartha, and Queen Elizabeth) scarce data can be found, and when data is available it is mostly outdated. This project aims to unveil these hidden habitats and provide critical insights into their ecological status by conducting a comprehensive assessment of ecological values, anthropogenic stressors, stakeholder engagement, and conservation and communication efforts. Freshwater ecosystems are globally vulnerable to multiple stressors, and national indicators show gaps and variability in monitoring coverage that justify targeted baseline work at sites such as Sucker Lake (Dudgeon et al., 2006; Environment and Climate Change Canada, 2024).

Main project objective

The principal aim of this project is to establish a comprehensive, long-term biomonitoring programme designed to evaluate the ecological health of the Sucker Lake site in Kawartha Highlands Provincial Park. This programme will facilitate ongoing preservation efforts and support sustainable management of the site by generating a robust baseline dataset for future comparisons. Upon TRCA choosing Sucker Lake as the main area for a field study (fig. 1), sampling will be done across an accessibility gradient, comparing accessible shoreline sectors with remote shoreline sectors within the same lake.

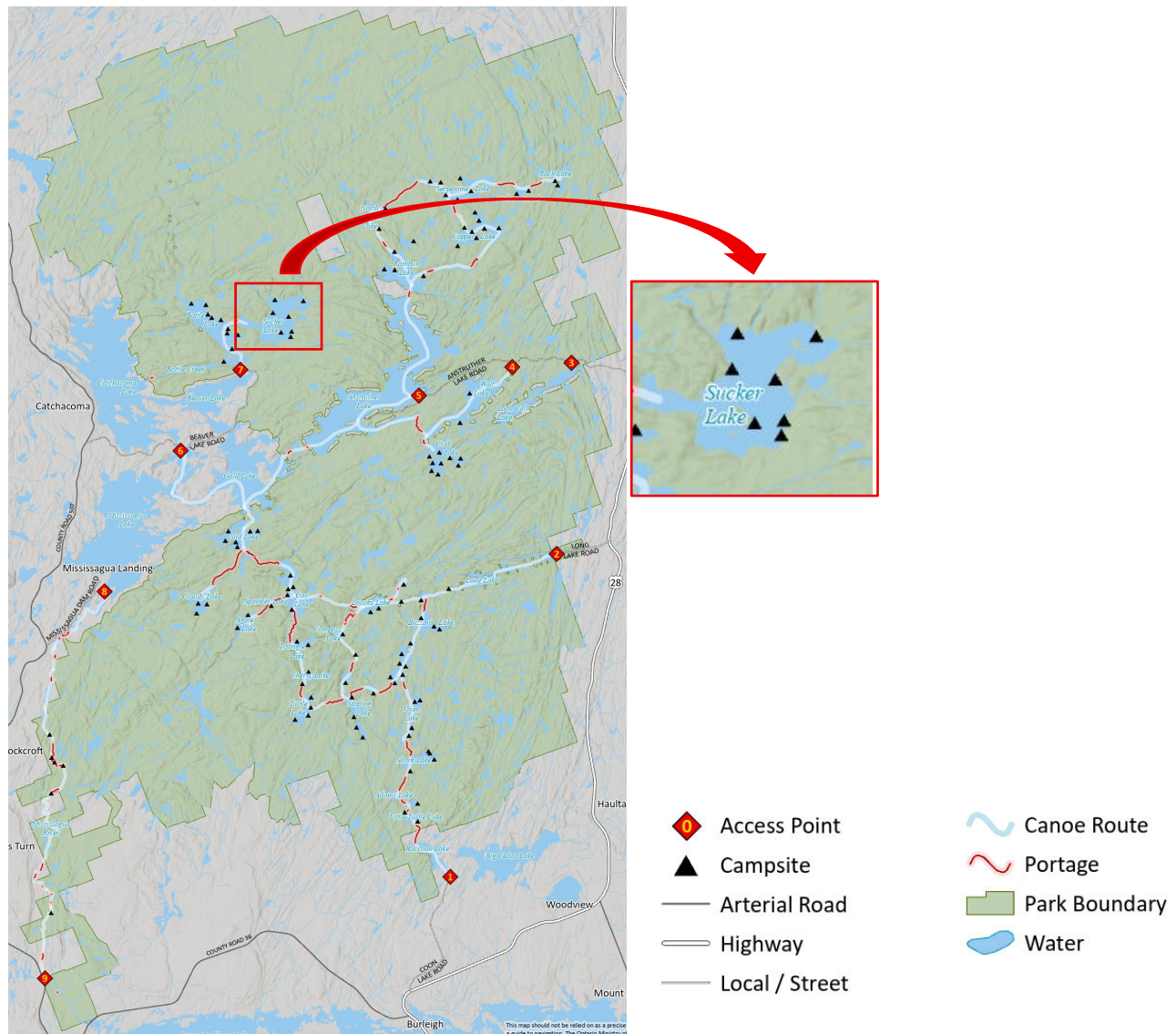


Figure 1 Kawartha Highlands Signature Site Park and chosen work site

Key focus areas

- Water quality assessment and benthic macroinvertebrate monitoring.
- Invasive plant monitoring for shoreline and shallow aquatic vegetation.
- Anthropogenic stressor evaluation across accessibility zones.

Objectives

Water monitoring objective

To initiate a long-term benthic macroinvertebrate biomonitoring program tailored to Sucker Lake, stratified by accessibility zones, to assess ecological health and establish a baseline dataset for future comparisons.

Specific objectives

- Identify and document benthic macroinvertebrate communities present across accessibility zones.
- Quantify water quality parameters to support interpretation of biological data.
- Compare biological and chemical indicators between accessible and remote shoreline sectors.

Invasive plant monitoring objective

To detect, map, and quantify invasive plant species in shoreline and shallow aquatic zones of Sucker Lake and to identify priority areas for management.

Anthropogenic stressor objective

To document the type and distribution of human use and shoreline modification across accessibility zones, and to relate those stressor indicators to biological and water quality data.

Materials and Methods

Study area and sampling design

Sucker Lake, Kawartha Highlands Park, will be the study site. The shoreline will be partitioned into two accessibility zones: accessible shorelines (adjacent to campsites, portages, or recreational access) and remote shorelines (minimal human access). In each zone 3 to 5 sampling sites will be established for each protocol, giving approximately 6 to 10 sites per protocol depending on logistics.

Map 1. Here I'll insert a map of the sites detailing the trail used to access the study zones points will be taken using GPS or phones (if we can get ArcGIS field maps account), campsites, and planned sampling locations. (started by drawing the outline of the lake using NAD83 / UTM Zone 17N as projection)

All field methods will follow provincial and national best practice where relevant, for comparability and QA/QC (Jones, Somers, & Reynoldson, 2007; Canadian Council of Ministers of the Environment, 2011).

1. Water quality assessment and benthic macroinvertebrate monitoring

Rationale and approach

Benthic macroinvertebrates provide integrative evidence of ecological condition while physical and chemical parameters provide mechanistic context. Combining both lines of evidence will produce a strong baseline for Sucker Lake (Hilsenhoff, 1988; Mandaville, 2002).

Field sampling

- Timing: planned field campaign on October 17–18, 2025.
- Site selection: 3–5 sites per zone (accessible, remote). Sites chosen to represent habitat variation along the shoreline.
- Water parameters measured in situ: temperature, dissolved oxygen, pH, conductivity, turbidity, and Secchi depth where feasible. Multiparameter meters will be calibrated daily according to manufacturer and APHA guidance (APHA, 2017).
- Nutrients: grab samples for total phosphorus, soluble reactive phosphorus, nitrate and ammonium will be collected in acid-washed bottles, preserved and submitted to an

accredited laboratory for analysis following CCME methods (Canadian Council of Ministers of the Environment, 2011).

- Benthic sampling: the travelling kick and sweep method using a 500 µm D-net to approximately 1 m depth. Two replicate samples per site. At each site 2 to 3 transects will be sampled to ensure adequacy (Jones et al., 2007). Record the time spent per transect to standardize effort.
- Preservation: samples preserved in 70% ethanol and transported to the lab.

Laboratory analysis

- Each replicate analyzed individually. Samples will be identified to family level and grouped to the OBBN 27 groups for metrics calculation (Jones et al., 2007; Mandaville, 2002).
- Subsampling via teaspoon method until a minimum of 100 individuals are counted per replicate. Senior TRCA taxonomist will verify identifications as quality assurance.

Data analysis

- Calculate Hilsenhoff Biotic Index, percent composition, Simpson's Diversity Index, and EPT percent. Interpret taxon tolerances per Mandaville (2002) and Hilsenhoff (1988). Compare indices between accessible and remote zones and test for differences using simple nonparametric tests given small pilot sample size.

2. Invasive plant monitoring

Rationale and target list

Monitoring will include terrestrial shorelines and nearshore macrophytes. Target terrestrial species are dog-strangling vine, garlic mustard, common buckthorn, Scots pine and *Phragmites* (fig. 2). Target aquatic and riparian species include Eurasian watermilfoil, curly-leaf pondweed and purple loosestrife (fig. 2). These species are known to occur and be invasive in Kawartha and surrounding regions and are relevant for shoreline habitat impact assessments (Ontario Invasive Plant Council, 2013; Kawartha Conservation, 2018).



Dog-strangling
vine



Garlic mustard



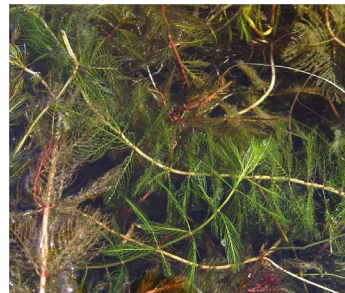
Common
buckthorn



Scots pine



Phragmites



Eurasian
watermilfoil



Curly-leaf
pondweed



Purple loosestrife

Figure 2 Pictures of the Invasive Species in Kawartha

Sampling design and methods

- At each shoreline site, establish a 100 m transect perpendicular to the disturbance gradient where applicable. Mark start and end with GPS.
- Place quadrats every 10 m: 1 × 1 m for herbaceous species, 5 × 5 m for shrubs and saplings. For aquatic macrophytes in ≤2 m depth, use rake tosses or visual quadrats along littoral transects to detect submerged and floating invasives. Methods align with the mapping and field forms in the Ontario invasive plant guidance (Ontario Invasive Plant Council, 2013).
- Record for each quadrat: species present, percent cover, density estimates, growth stage, and evidence of native vegetation suppression. Photograph each quadrat and note GPS coordinates.
- For patches > 5 m² walk the perimeter with GPS to create polygon features for mapping.

Frequency

- During this initial campaign a single seasonal survey will be completed, with recommended repeat visits in spring and mid-summer for detection reliability and phenology differences. The Ontario guide and provincial practice recommend multiple seasonal visits for some species (Ontario Invasive Plant Council, 2013).

Map 2: here we'll Invasive transects we chose and preliminary polygons. We can also include photo point locations to help future monitoring

3. Anthropogenic stressor evaluation

Rationale

Assessing human use indicators along the same sampling sites enables direct comparison between use intensity and ecological metrics. Recreational impacts on nearshore vegetation and water quality are well documented, and a gradient approach within a single lake can reveal localized effects without confounding lake-to-lake variability (Dustin et al., 2017; Schafft et al., 2021).

Methods

- Use the same shoreline sites. At each site record shoreline type (natural, altered), structures, evidence of erosion, litter, fire rings, number and condition of campsites, and visible trails. Use a standardized checklist for consistency.
- Count boats and visible users during fixed observation periods. Photograph disturbances and log GPS coordinates.
- Collect nearshore water samples at ~0.5 m depth for supporting chemistry and turbidity measures. Collect one mid-lake grab for comparison where safe and feasible.
- Sediment: where relevant, collect a top 5 cm sediment grab at selected nearshore locations for later laboratory analysis of organic content and contaminants if budget allows.

QA/QC and safety

- Duplicate field sheets, field blanks for lab samples, calibration logs for instruments, and 10% duplicate samples for lab analyses in line with CCME and APHA recommendations (Canadian Council of Ministers of the Environment, 2011; APHA, 2017).
- Two person teams, Personal floating devices during boat work, daily check-ins, and a field Emergency Action Plan. TRCA risk approvals will be confirmed before dispatch.

Materials and Equipment

The following table summarizes the equipment and materials required for the fieldwork components of this project, including water quality, benthic invertebrate, and invasive plant monitoring. Items were selected to ensure consistency with provincial field standards (OBBN, OIPC) while maintaining a cost-efficient approach suitable for small field teams.

Table 1 Required Materials and Equipment for Sucker Lake Fieldwork

Category	Item	Quantity	Notes / Purpose
General Field Gear	Boat or canoe (available)	1	Used for accessing sampling sites across Sucker Lake.
	Life jackets / PFDs	As needed	Required for all water-based work.
	First aid kit	1	Standard safety equipment.
	Gloves (medium)	2 packs	Protects hands during sampling and specimen handling.
	Trash bags	2	Used for waste collection and site cleanup.
Water Quality Monitoring	YSI multiparameter probe	1	Measures temperature, pH, DO, and conductivity.
	Sample buckets (white)	3	Used to collect and transfer water samples.
	Distilled water (4 L)	1	Used for rinsing instruments.
	Wash bottle	1	Used to clean sampling equipment.
	Waterproof field notebook	1	For recording site and parameter data.
Benthic Invertebrate Sampling	500 µm D-frame kick net	1	For collecting benthic samples from littoral zones.
	500 µm metal sieve	1	Used to separate fine sediment from invertebrates.
	Sample jars (500 mL)	20	For sample storage and labeling.
	70% Ethanol (4 L)	2	For sample preservation.
	Forceps, droppers	1 each	For sorting and transferring specimens.

	Dissecting microscope	1	For laboratory identification of taxa.
	Identification keys (OBBN) & taxonomic guides	1 set	Used to identify organisms to family level.
	Sorting trays / Petri dishes	5	Used for visual separation of specimens.
	Labelled vials	2	For preserved subsamples.
Invasive Aquatic Plant Monitoring	Measuring tape (100 m)	1	Used to mark transects and estimate vegetation cover.
	Flagging tape (colored)	1 roll	Marks transect start/end points.
	Clipboard & waterproof datasheets	1	For data recording along transects.
	Smartphone GPS app (Avenza, GPS Essentials)	1	Used to record coordinates of transects and photo points.
	Camera or phone	1	Used to document invasive species presence and extent.
Sample Handling & Transport	Sample transport containers	3	To safely move up to 20 sampling jars.
	Labels and waterproof markers	2 sets	Used to mark samples with site codes and dates.

Expected outcomes

Water / benthic monitoring outcomes

- A baseline dataset of benthic macroinvertebrate composition for Sucker Lake, including HBI, EPT percent, Simpson's Diversity and other metrics.
- A paired set of water chemistry data for each biological sampling site to support interpretation.
- Comparative analysis of accessible versus remote shoreline sectors to detect relationships between human accessibility and ecological condition.

Invasive plant monitoring outcomes

- Baseline maps showing occurrence, percent cover and density of target terrestrial and aquatic invasive plants.

- Identification of priority management polygons for follow up control or monitoring.
- Recommendations on seasonal timing for follow up surveys based on detection success.

Anthropogenic stressor outcomes

- A quantified inventory of shoreline modifications, campsite impacts, litter, and visitor use at sampled sites.
- Integration of stressor indicators with water and biological data to inform management options, such as visitor education, campsite relocation, or shoreline restoration.

Cross-protocol outcomes

- A validated pilot monitoring package that TRCA and NYGEP can scale in future years or across nearby lakes.
- Materials for community communication and education summarizing key findings and actions.

Limitations and recommended refinements

- This single seasonal campaign will provide a snapshot baseline. For robust trend detection, multi-season and multi-year repeats are recommended (Canadian Council of Ministers of the Environment, 2011).
- Detection probability for some aquatic invasives is seasonal. Plan a spring and mid-summer revisit if resources allow (Ontario Invasive Plant Council, 2013).

Table 2 Next Monitoring Windows and Recommended Frequency for Key Protocols

Protocol	Next Monitoring Window	Recommended Frequency	Justification
Water Quality	May–June 2026 (spring) and August 2026 (late summer)	Twice per year	Captures seasonal variation in nutrient levels, dissolved oxygen, and temperature changes (APHA, 2017; CCME, 2011)
Invasive Species	July and September 2026	Annually during growing season	Early and late season observations help detect emergence and spread of aquatic invasive plants (Ontario Invasive Plant Council, 2013)
Anthropogenic Pressures	July–August 2026	Every 1–2 years	Reflects changes in recreation intensity, shoreline modification, and litter accumulation patterns (Ontario Parks, 2008)

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